Institute of Engineering and Technology, Lucknow

Evaluation scheme & Syllabus

For

M. Tech.

In

Power& Energy System

Based on

CHOICE BASED CREDIT SYSTEM (CBCS)

(Effective from the session:2020-21)

Institute of Engineering and Technology, Lucknow

(Constituent Institute of DR.A.P.J. Abdul Kalam Technical University, Lucknow)

M.Tech : Power& Energy Systems

Semester 1												
S. No.	Subject Code	Name of Subject	Periods				Evaluation Scheme					
							Theory			Practical		Subject Total
			L	T	P	Credit	CT	TA	ESE	TA	ESE	
1	MTPE-101	Advanced Power System Analysis	3	0	0	3	20	10	70	-	-	100
2	MTPE-102	Solar Energy Conversion	3	0	0	3	20	10	70	-	-	100
3	MTPE-011 - MTPE-013	Departmental Elective 1	3	0	0	3	20	10	70	-	-	100
4	MTPE-021MTPE-023	Departmental Elective 2	3	0	0	3	20	10	70	-	-	100
5	RM-101	Research Process & Methodology	3	0	0	3	20	10	70	-	-	100
6	MTPE-151	Power System Simulation Lab	0	0	3	2	-	-	-	20	30	50
7	MTPE-152	Renewable Energy Lab	0	0	2	2	-	-	-	20	30	50
TOTAL						18						600

Semester 2												
S. No.	Subject Code	Name of Subject	Periods				Evaluation Scheme					
							Theory			Practical		Subject Total
			L	T	P	Credit	CT	TA	ESE	TA	ESE	
1	MTPE-201	Advanced Power Electronics	3	0	0	3	20	10	70	-	-	100
2	MTPE-202	Wind Energy Conversion System	3	0	0	3	20	10	70	-	-	100
3	MTPE-031 - MTPE-033	Departmental Elective 3	3	0	0	3	20	10	70	-	-	100
4	MTPE-041 - MTPE-043	Departmental Elective 4	3	0	0	3	20	10	70	-	-	100
5	MTPE-051 - MTPE-053	Departmental Elective 5	3	0	0	3	20	10	70	-	-	100
5	MTPE-251	Power Electronics Lab	0	0	3	2	-	-	-	20	30	50
7	MTPE-252	Seminar 1	0	0	2	2	-	-	-	50	-	50
TOTAL						18						600

Semester 3												
S. No.	Subject Code	Name of Subject	Periods				Evaluation Scheme					
							Theory			Practical		Subject Total
			L	T	P	Credit	CT	TA	ESE	TA	ESE	
1	MTEE-351	SEMINAR 2	0	0	6	3	-	-	-	100	-	**100**
2	MTEE-352	DISSERTATION	0	0	30	15	-	-	-	200	300	**500**
TOTAL						**18**						**600**

Semester 4												
S. No.	Subject Code	Name of Subject	Periods				Evaluation Scheme					
							Theory			Practical		Subject Total
			L	T	P	Credit	CT	TA	ESE	TA	ESE	
1	MTPE-451	Dissertation (Final)	0	0	36	18				200	400	600
TOTAL						**18**	**600**					

Departmental Elective- I	
MTPE-011	Power System Optimization
MTPE-012	Power System Reliability and Planning
MTPE-013	Policies and Economics of International Energy

Departmental Elective- II	
MTPE-021	Electric Power Distribution System
MTPE-022	Advanced Electric Drives
MTPE-023	Power Quality

Departmental Elective- III	
MTPE-031	FACTS and Custom Power Devices-Application to Power System
MTPE-032	Electric Vehicles

Departmental Elective- III	
MTPE-033	Power System Deregulation

Departmental Elective- IV	
MTPE-041	Smart grid
MTPE-042	Distributed Generation
MTPE-043	Machine Learning Applications to Power System

Departmental Elective- V	
MTPE-051	Power System Dynamics & Control
MTPE-052	Modern Power system Protection
MTPE-053	Synchrophasors- Applications to Power System

MTPE 101 -ADVANCED POWER SYSTEM ANALYSIS

L T P 3 0 0

Course Outcomes:

CO-1- Elucidate different types of mathematical models for analysis.
CO-2- Understand the different methodologies of load flow studies for the power network.
CO-3-To impart knowledge on system contingencies and behavior of system to ensure security for the system operation.
CO-4- Analyze the formation of suitable mathematical model for obtaining optimal performance during operational conditions.
CO-5-To demonstrate different numerical integration methods and factors influencing transient stability.

UNIT I

Network matrix: Physical interpretation of bus admittance and impedance matrices, introduction to admittance matrix formulation, formation of admittance matrix due to inclusion of regulating transformer, development of admittance matrix using singular transformation, modification of admittance matrix for branch addition/ deletion.

UNIT II

Complex power flow: Analytical formulation of complex power flow solution, Gauss-Seidal method of power flow, Newton Raphson method of power flow, algorithm for solving power flow problem using N-R method in rectangular form, algorithm for solving power flow problem using N-R method in polar form, fast decoupled load flow method.

UNIT III

Power System Stability: Definitions, classification of stability-rotor angle and voltage stability, synchronous machine representation for stability study. Transient stability: Assumptions for transient stability, derivation of swing equation, swing equation for synchronous machine connected to infinite bus, swing equation for a two machine system.

UNIT IV

Solution of swing equation by Euler and RungeKutta method, equal area criterion, critical clearing angle, application of critical clearing angle to transient stability of synchronous machine. Methods of improving transient stability: reducing fault clearance time, automatic reclosing, single phase reclosing, electric braking, voltage regulators, fast governor action, high speed excitation system.

UNIT V

Voltage stability: Definition and classification of voltage stability, mechanism of voltage collapse, analytical concept of voltage stability for a two bus system, expression for critical receiving end voltage and critical power angle at voltage stability limit for a two bus power system, PV and QV curves, L index for the assessment of voltage stability.

REFRENCES:

1. J.J Grainger&W.D. Stevenson, “Power System Analysis”, McGraw Hill,2003
2. A.R. Bergen & Vijay Vittal, “Power system analysis”,Pearson,2000, 2.
3. L.P Singh, “Advanced Power System Analysis and Dynamics”, NewAge International, 2006.

4. G.L. Kausic, “Computer Aided Power System Analysis”, Prentice Hall India, 1986.
5. A.J. Wood, “Power generation. Operation and control”, John Wiley,1994.
6. P.M Anderson, “Faulted Power System analysis”, IEEE Press,1995.
7. Power system Analysis by Charles A: Gross John Wiley &Sons.

MTPE 102-SOLAR ENERGY CONVERSION

L T P 3 0 0

Course Outcomes

CO- 1-Demonstrate the working principle of various solar energy systems.
CO-2-Analyze the properties of solar energy resource, PV and ST system operation and component specifications.
CO-3-Compute circuit parameters, solar geometry and cell/array performance parameters.
CO-4-Design of PV systems for domestic, commercial and industrial applications.
CO-5-Familiarize with solar energy policies, costing and PV system designing software.

UNIT I

INTRODUCTION:

Trends in energy consumption-world energy scenario, energy resources and their availability, conventional and renewable sources, need to development new energy technologies.

UNIT II

SOLAR ENERGY FUNDAMENTALS:

The Sun as a Source of Energy, The Earth, Sun and Earth Radiation Spectrums. Extra-terrestrial and Terrestrial Radiations, Spectral Energy Distribution of Solar Radiation, Depletion of Solar Radiation, Measurements of Solar Radiation, Solar Time (Local Apparent Time), Solar Radiation Geometry, Solar Day Length, Empirical Equations for Estimating Solar Radiation, Solar radiation on Horizontal Surface, Solar Radiation on Inclined Plane Surface.

UNIT III

SOLAR THERMAL ENERGY CONVERSION:

Solar Collectors, their classification and working. Solar Water Heater, Solar Passive Space-Heating, and Cooling Systems, Solar Industrial Heating Systems, Solar Thermal Water Pump, Solar Pond Electric Power Plant, Distributed Collector Soar Thermal Electric Power Plant, Central Tower Receiver Power Plant.

UNIT IV

PHOTOVOLTAIC ENERGY CONVERSION:

Semiconductors, photoconduction, solar cell and their characteristics, influence of insolation and temperature, classification of Solar Cells, PV arrays, effect of cell-mismatch and shadowing, maximum power point tracking, MPPT algorithms.

UNIT V

POWER CONDITIONING SCHEMES:

Electrical storage with batteries, switching devices for solar energy conversion, DC power conditioning converters, AC power conditioners, synchronised operation with grid supply, harmonic problem. Solar

Power Systems (Grid Connected, Standalone etc.)

REFRENCES:

1. Mukund R Patel,"Wind and Solar Power Systems", CRC press, 2004
2. Rai G D,"Wind Power Energy Resources", Khanna Publishers New Delhi 2002
3. Daniel Hunt V," Wind Power - A Hand Book of WECS", VAN NOSTREND CO.
4. Thomas Markvart and Luis Castaser," Practical Hand Book of Photovoltaics Elsevier Publications, UK ,200
5 .A.Duffie& W.A. Beckman:“Solar Engineering of Thermal Process”, 4thEdition, Wiley
6. S.A.Kalogirou: Solar Energy Engineering, Academic Press, 2013.

MTPE 011- POWER SYSTEM OPTIMIZATION TECHNIQUES

L T P 3 0 0

Course Outcomes

CO-1-Identify different types of optimization techniques and problems.
CO-2-Demonstrate the techniques like calculus based classical optimization, linear programming, nonlinear programming, dynamic programming.
CO-3- Analyze the principles and techniques described in second outcome to solve sample mathematical and practical optimization problems.
CO-4- Measure and evaluate the performance obtained by applying optimization techniques to solve mathematical programming problems.
CO-5- Apply ANN, fuzzy logic and optimization algorithms to solve the problems.

UNIT –I

Introduction to optimization and classical optimization techniques Linear Programming: Standard form, geometry of LPP, Simplex Method pf solving LPP, revised simplex method, duality, decomposition principle, and transportation problem.

UNIT –II

Non-Linear Problem (NLP): One dimensional methods, Elimination methods, Interpolation methods, Unconstrained optimization Techniques-Direct search and Descent methods, constrained optimization techniques, direct and indirect methods.

UNIT –III

Dynamic Programming: Multistage decision processes, concept of sub-optimization and principle of optimality, conversion of final value problem into an initial value problem CPM and PERT.

UNIT –IV

Genetic Algorithm: Introduction to genetic Algorithm, working principle, coding of variables, fitness function. GA operators; Similarities and difference s between Gas and traditional methods; Unconstrained and constrained optimization using Genetic Algorithm, real coded gas, Advanced Gas, global optimization using GA.

UNIT –V

Applications to Power system: Economic Load Dispatch in thermal and Hydro-thermal system using GA and classical optimization techniques, Unit commitment problem, reactive power optimization. Optimal power flow, LPP and NLP techniques to optimal flow problems.

REFERENCES:

1. Rao S. S “Optimization: Theory and Application Wiley Eastern Press”, 2nd edition 1984.
2. Taha H.A., Operations Research –An Introduction, Prentice Hall of India, 2003.
3. Fox, R.L., „Optimization methods for Engineering Design, Addition Wiley, 1971.
4. K. Deb, “Multi-objective Optimization using Evolutionary Algorithms”, Wiley, 2002

MTPE 012-POWER SYSTEM RELIABILITY & PLANNING

L T P 3 0 0

Course Outcomes:

CO-1- Analyze the reliability analysis applied to power systems.

CO-2- Apply the Markov Chains and application to power systems.

CO-3- Measure and evaluate stability analysis of generation systems.

CO-4- Analyze the decomposition techniques applied to power system.

CO-5- Evaluate the peak demand and energy requirements of system using forecasting techniques.

UNIT I

SYSTEM PLANNING:

Introduction, Objectives & Factors affecting to System Planning, Short Term Planning, Medium Term Planning, Long Term Planning, Reactive Power Planning.

Load forecasting: Classification and characteristics of loads. Approaches to load forecasting. Forecasting methodology. Energy forecasting.

UNIT II

RELIABILITY:

Reliability, Failure, Concepts of Probability, Evaluation Techniques (i) Markov Process (ii) Recursive Technique, Stochastic Prediction of Frequency and Duration of Long & Short Interruption, Adequacy of Reliability, Reliability Cost.

Basic Reliability Concepts: General reliability function, Markov Chains and processes and their applications, simple series and parallel system models.

UNIT III

GENERATION PLANNING AND RELIABILITY:

Generation Sources, Integrated Resource Planning, Generation System Model, Loss of Load (Calculation and Approaches), Outage Rate, Capacity Expansion, Scheduled Outage, Loss of Energy, Evaluation Methods, Interconnected System, Factors Affecting Interconnection under Emergency Assistance

UNIT IV

TRANSMISSION PLANNING AND RELIABILITY:

Introduction, Objectives of Transmission Planning, Network Reconfiguration, System and Load Point Indices, Data required for Composite System Reliability.

Transmission System Reliability Evaluation: Average interruption rate method. The frequency and duration method. Stormy and normal weather effects.

UNIT V

DISTRIBUTION PLANNING AND RELIABILITY:

Radial Networks, Network Reconfiguration, Evaluation Techniques, Interruption Indices, Effects of

Lateral Distribution Protection, Effects of Disconnects, Effects of Protection Failure, Effects of Transferring Loads, Distribution Reliability Indices, Parallel & Meshed Networks, Bus Bar Failure, Scheduled Maintenance, Temporary and Transient Failure, Breaker Failure.

REFERENCES:

1. R.L. Sullivan "Power System Planning", Tata McGraw Hill Publishing Company Ltd.
2. Roy Billinton & Ronald N. Allan "Reliability Evaluation of Power System", Springer Publication
3. T. W. Berrie "Electricity Economics & Planning", Peter Peregrinus Ltd., London.
4. Dasari, S., Electric Power System Planning, IBT Publishers (1999).
5. TuranGonen, "Electric Power Distribution system Engineering", Second edition, Taylor & Francis
,2007.

MTPE 013 -POLICIES AND ECONOMICS OF INTERNATIONAL ENERGY

L T P 3 0 0

Course Outcomes:

CO-1- Understand the basics of the Indian and World Energy Scenario and world energy use resources, Energy cycle on earth etc.
CO-2- Describe the types of energy , energy storage and energy conversion systems.
CO-3- Illustrate the energy economy final energy consumption energy needs of growing economy.
CO-4- Describe the Energy conservation act its features and related policies other acts. The integrated energy policy 2006 etc.
CO-5- Identify the factors related energy and environment, air pollution climate changes and its impacts on sustainable development

UNIT I

Role of energy in economic development and social transformation: Energy & GDP, GNP and its dynamics. Discovery of various energy sources: Energy Sources and Overall Energy demand and availability, Energy Consumption in various sectors and its changing pattern, Exponential increase in energy consumption and Projected future demands. Energy Resources: Coal, Oil, Natural Gas, Nuclear Power and Hydroelectricity, Solar and Other Renewable etc. Depletion of energy sources and impact exponential rise in energy consumption on economies of countries and on international relations.

UNIT II

Chemical and Nuclear: Non Proliferation, Energy Security, Energy Consumption and its impact on environmental climatic change. International Energy Policies of G-8 Countries, G-20 Countries, OPEC Countries, EUCountries. International Energy Treaties (Rio, Montreal, Kyoto), INDO-US Nuclear Deal. Future Energy Options: Sustainable Development, Energy Crisis: Transition from carbon rich and nuclear to carbon free technologies, parameters of transition.

UNIT III

Energy resources & Consumption: Commercial and non-commercial forms of energy, Fossil fuels, Renewable sources including Bio-fuels in India, their utilization pattern in the past, present and future

projections of consumption pattern, Sector wise energy consumption Impact of Energy on Economy, Development and Environment, Energy for Sustainable Development, Energy and Environmental policies, Need for use of new and renewable energy sources.

UNIT IV

Status of Nuclear and Renewable Energy: Present Status and future promise Energy Policy Issues: Fossil Fuels, Renewable Energy, Power sector reforms, restructuring of energy supply sector, energy strategy for future. Energy Conservation Act-2001 & its features, Electricity Act-2003 & its features. Framework of Central Electricity Authority (CEA), Central & States Electricity Regulatory Commissions (CERC & ERCs)

UNIT V

Energy Policy:Global Energy Issues, National & State Level Energy Issues, National & State Energy Policy, Industrial Energy Policy, Energy Security, Energy Vision. Energy Pricing & Impact of Global Variations. Energy Productivity (National & Sector wise productivity).

REFERENCES:

1. Energy for a sustainable world: Jose Goldenberg, Thomas Johansson, A.K.N.Reddy, Robert Williams (Wiley Eastern).
2. Energy policy for : B.V. Desai (Wiley Eastern),
3. Modelling approach to long term demand and energy implication: J.K.Parikh.
4. Energy Policy and Planning: B.Bukhootsow.
5. TEDDY Year Book Published by Tata Energy Research Institute (TERI),
6. World Energy Resources: Charles E. Brown, Springer2002.
7. International Energy Outlook' -EIA annual Publication
8. Heat and Thermodynamics – M.W. Zemansky (McGraw Hill Publication
9. Principles of Energy Conversion: A.W. Culp (McGraw Hill International edition.)
10. BEE Reference book: no.1/2/3/4.

MTPE 021 -ELECTRIC POWER DISTRIBUTION SYSTEM

L T P 3 0 0

Course Outcomes

CO-1- Understand significance of of the power distribution system.
CO-2- Illustrate the distribution automation and its application in practice.
CO-3-To comprehend the SCADA system.
CO-4-Design the protective systems and co-ordinate the devices.
CO-5- Analyze the capacitive compensation.

UNIT I

Distribution of Power, Management, Power Loads, Load Forecasting Short-term & Long-term, Power System Loading, Technological Forecasting.

UNIT II

Advantages of Distribution Management System (D.M.S.) Distribution Automation: Definition, Restoration / Reconfiguration of Distribution Network, Different Methods and Constraints, Power Factor Correction.

UNIT III

Interconnection of Distribution, Control & Communication Systems, Remote Metering, Automatic Meter Reading and its implementation.

SCADA: Introduction, Block Diagram, SCADA Applied To Distribution Automation. Common Functions of SCADA, Advantages of Distribution Automation through SCADA.

UNIT IV

Calculation of Optimum Number of Switches, Capacitors, Optimum Switching Device Placement in Radial, Distribution Systems, Sectionalizing Switches – Types, Benefits, Bellman's Optimality Principle, Remote Terminal Units, Energy efficiency in electrical distribution & Monitoring.

UNIT V

Maintenance of Automated Distribution Systems, Difficulties in Implementing Distribution. Automation in Actual Practice, Urban/Rural Distribution, Energy Management, AI techniques applied to Distribution Automation.

REFERENCES:

1. A.S. Pabla, "Electric Power Distribution", Tata McGraw Hill Publishing Co. Ltd., Fourth Edition.
2. M.K. Khedkar, G.M. Dhole, "A Text Book of Electrical power Distribution Automation", University Science Press, New Delhi.
3. Anthony J Panseni, "Electrical Distribution Engineering", CRC Press.
4. James Momoh, "Electric Power Distribution, automation, protection & control", CRC Press

MTPE 022- ADVANCD ELECTRIC DRIVES

L T P 3 0 0

Course Outcomes:

CO-1-Comprehend state of the art technology of dc and ac advanced drives.
CO-2-Analyze performance of dc and ac drives.
CO-3-To identify suitable drives according to the application.
CO-4-Design the advanced drive and compare the performance with the existing one.
CO-5-To estimate & solve harmonics and power factor related problems in controlling AC and DC drives.

UNIT I

Review of electric motors & Solid state converters: Speed control techniques of DC, Induction & synchronous motor, Converters, inverters, chopper and cyclo converter operation, Effects of power electronic equipment's on load side & supply side.

Review of closed loop controllers, sensors &transducers: PI, PID, Variable structure. AC, DC & Pulse tacho-generators.

UNIT II

DC Drives: Converter & chopper fed DC drive, Reversing, Starting, Regenerativebreaking,four quadrant operation, and High power application.

UNIT III

AC Drive: Inverter & cyclo converter fed drive, Vector control, Sensor less operation, Linear electrical motor concept, Synchronous motor drive.

UNIT IV

SpecialDrives: Switched reluctance& permanent magnet brushless DC Operation, Converters, Characteristics & Control, PLC based drives.

Servo drives & stepper motor- AC & DC Servomotor, Stepper motor, Control techniques, Controllers,Micro stepping, Sensor less operation.

UNIT V

Power Quality & energy Conservation- Line Side pollution, standards, Harmonic elimination techniques in converter, Filters, Energy efficient electric motors, Pay back periods, Energy conservation through soldstate control.

REFERENCES:

1. Ned Mohan, T.M. Undeland, W.P. Robbins, Power Electronics-Converters, Applications anddesign", John Wiley & Sons.
2. J.M.D. Murphy, F.O. Turnbull, "Power Electronic Control of AC motors", Pergamon Press.
3. P.C. Sen, D.C. drive, Pergamon Press
4. B.K. Bose, Power Electronics & AC drive prentice Hall.
5. Dubey G.K. "Power semi-Conductor controller drives, Prentice Hall.
6. VedamSubramanyam, "Electrical Drives".
7. T.J.E. Miller, Switched Reluctance & P.M. B.L. DC motor, Pergamon Press

MTPE 023-POWER QUALITY

L T P 3 0 0

Course Outcomes:

CO-1-Identify the issues related to power quality in power systems.
CO-2-Address the problems of transient and long duration voltage variations in power systems.
CO-3- Analyze the effects of harmonics and study of different mitigation techniques.
CO-4- Identify the importance of custom power devices and their applications.
CO-5- Acquire knowledge on different compensation techniques to minimize power quality disturbances.

UNIT I

Electric power quality phenomena: - Impacts of power quality problems on end users, Power quality standards, power quality monitoring.

Power quality disturbances: - transients, short duration voltage variations, long duration voltage variations, voltage imbalance, wave-form distortions, voltage fluctuations, power frequency variations, power acceptability curves.

UNIT II

Power quality problems: poor load power factor, loads containing harmonics, notching in load voltage, dc offset in loads, unbalanced loads, disturbances in supply voltage.

UNIT III

Transients: Origin and classification- capacitor switching transient-lighting-load switching-impact on users-protection mitigation.

UNIT IV

Harmonics: harmonic distortion standards, power system quantities under non sinusoidal conditions-harmonic indices-source of harmonics-system response characteristics-effects of harmonic distortion on power system apparatus –principles for controlling harmonics, reducing harmonic currents in loads, filtering, modifying the system frequency response- Devices for controlling harmonic distortion, inline reactors or chokes, zigzag transformers, passive filters, active filters.

UNIT V

Power quality conditioners: Shunt and series compensators, DSTATCOM-dynamic voltage restorer, Unified Power Quality Conditioners (UPQC).

REFERENCES:

1. Ghosh Arindam and Ledwich Gerard, 'Power quality enhancement using custom power devices' Springer.
2. Arrillaga J., Watson N. R. and Chen S., 'Power System Quality Assessment' Wiley.
3. Caramia P, Carpinelli G and Verde P, 'Power quality indices in liberalized markets' – Wiley .
4. Angelo Baggini 'Handbook of Power Quality' – Wiley.
5. . Roger C. Dugan, Mark F. Mcgranaghan, Surya Santoso, "Electrical Power System Quality", McGraw Hill
6. C. Sankaran, Power Quality CRC Press, USA 3. Wilson E. Kazibwe, "Electrical Power Quality Control Techniques", Van Nostrand Reinhold.

MTPE151-POWER SYSTEM SIMULATION LAB

L T P 0 0 3

Course Outcomes:

CO-1-To analyze load flow solution obtained using GS and NR methods.
CO-2- Elucidate different types of symmetrical and unsymmetrical faults.
CO-3-To analyze transient stability and load frequency deviation in single and two area systems.
CO-4- Calculate the steady-state power flow in a power system.
CO-5- Determine optimal power generation & losses of a power system.

LIST OF EXPERIMENTS:

1. Formation of Y-BUS matrix using MATLAB coding.
2. MATLAB coding N-R load flow in polar coordinates.
3. Load flow calculation using MATLAB and PST package.
4. Optimal power flow using PSAT.
5. Distribution load flow.
6. Symmetrical and unsymmetrical fault studies.
7. Voltage Instability Analysis.
8. Harmonic Analysis.
9. Economic Load Dispatch with Hydro thermal power plants.
10. Simulation of Facts controllers
11. Simulation of single -area and Two -area Systems.

NOTE- Minimum of 10 experiments are to be conducted.

MTPE152-RENEWABLE ENERGY LAB

L T P 0 0 3

Course Outcomes:

CO-1- Describe the simulation tools MATLAB/Simulink, PS pice, PSCAD, for solving Electrical engineering problems related to renewable energy.

CO-2-Implement and verify control strategies in the simulation platform for renewable energy applications.

CO-3- Measure and evaluate the solar radiations and test the performance of different solar thermal applications.

CO-4- Characterize solar cells and analyze different parameters such as power flow, efficiency of different components such PV module, battery, inverter and PV system.

CO-5- Characterize the properties of solid hydrogen fuel cells along with performance testing.

LIST OF EXPERIMENTS:

1. To determine the efficiency of Solar PV panel at different irradiance levels.
2. To determine the efficiency of a wind turbine for different wind speeds
3. Test the Capabilities of the Hydrogen Fuel Cells and Capacitors
4. Effect of Temperature on Solar Panel Output
5. Variables Affecting Solar Panel Output
6. Effect of Load on Solar Panel Output
7. Wind Turbine Output: The Effect of Load
8. Test the Capabilities of Solar Panels and Wind Turbines

MTPE 201-ADVANCE POWER ELECTRONICS

L T P 3 0 0

Course Outcomes:

CO-1- Analyze the different dc-dc voltage regulators.

CO-2-Simulate and analyze resonant converters.

CO-3-Identify the appropriate phase shifting converter for a multi-pulse converter

CO-4- Analyze various multi-level inverter configurations.

CO-5- Recognize the role of power electronics in the improvement of energy usage efficiency and the applications of power electronics in emerging areas.

Unit-I: Switching Voltage Regulators:

Introduction; Linear power supply (voltage regulators); Switching voltage regulators; Review of basic dc-dc voltage regulator configurations -Buck, Boost, Buck-Boost converters and their analysis for continuous and discontinuous mode; Other converter configurations like Flyback converter, Forward converter, Half bridge, Full bridge configurations, Push-pull converter, C'uk converter, Sepic Converter; Design criteria for SMPS; Multi-output switch mode regulator.

Unit-II: Resonant Converters:

Introduction, need of resonant converters, Classification of resonant converters, Load resonant converters, Resonant switch converters, zero voltage switching dc-dc converters, zero current switching dc-dc converters, clamped voltage topologies.

Unit-III: Multi-Level Converters:

Need for multi-level inverters, Concept of multi-level, Topologies for multi-level: Diode Clamped, Flying capacitor and Cascaded H-bridge multilevel Converters configurations; Features and relative comparison of these configurations applications, Introduction to carrier based PWM technique for multi-level converters.

Unit-IV: Multipulse Converters:

Concept of multi-pulse, Configurations for m-pulse (m=12,18,24 ….) converters, Different phase shifting transformer (Y-Δ1, Y-Δ2, Y-Z1 and Y-Z2) configurations for multi-pulse converters, Applications.

Unit-V: Hvdc Transmission:

Introduction, Operation of 12-pulse converter as receiving and sending terminals of HVDC system, Equipment required for HVDC System and their significance, Comparison of AC and DC transmission, Control of HVDC transmission.

REFERENCES:

1. Ned Mohan, Tore M. Undeland and William P. Robbins, "Power Electronics – Converters, Applications and Design", John Willey & sons, Inc., 3rd ed., 2003.
2. Muhammad H. Rashid, "Power Electronics - Circuits, Devices and Applications", Prentice Hall of India, 3rd ed., 2009.
3. . Bin Wu, "High Power Converters and AC Drives", John Willey & sons, Inc., 2006.
4. Derek A. Paice "Power Electronic Converter Harmonics – Multipulse Methods for Clean Power", IEEE Press, 1996. 5. Muhammad H. Rashid , "Power Electronics Handbook", Elsevier, 3rd ed., 2011.

5. P.C.Sen, “Modern Power Electronics ”, S. Chand and Co. Ltd., New Delhi, 2000.
6. Vijay K. Sood, “HVDC and FACTS Controllers Applications of Static Converters in Power Systems”, Kluwer Academic Publishers, Boston, 2004.
7. L. Umanand, “Power Electronics Essentials and Applications”, Wiley India Ltd., 2009

MTPE 202-WIND ENERGY CONVERSION SYSTEM

L T P 3 0 0

Course Outcomes:

CO1: Describe the working principle and basic components of wind energy conversion.
CO2: Ability to design power electronic converters for wind systems.
CO3: Ability to design power electronic converters for grid connected wind systems.
CO4: Elucidate different types of power electronic converters with renewable energy sources.
CO5: To develop various MPPT techniques for wind systems.

Unit-I: Introduction to wind power and The Wind Resource

Introduction to Wind Power Plants, Historical Background, Global Wind Power Growth, Indian Wind Power Growth, Drivers and Bottlenecks for Wind Power Development, Strengths and Limitations of Wind Power, Types of Wind, Wind Profiling, Turbulence, Hill and Tunnel Effect, Energy in the Wind, Energy Production, Weibull and Rayleigh Distribution, Wind Rose, Wind Power Density, Energy and Power, Energy Pattern Factor, Siting.

Unit-II: Wind Energy Conversion

Introduction to Wind Resource Assessment (Anemometers, SODAR, LIDAR), Construction of the Wind Power Plant, Rotation Principle, Drag Principle, Lift Principle, Forces on a Rotor Blade, Factors Affecting Performance of Rotor, Thrust and Torque on Rotor, Power Curve.

Wind Turbine Aerodynamics

Aerodynamic Power Regulation, Stall Controlled WPP, Pitch Controlled WPP, Active-Stall Controlled WPP, Halting a WPP, Other Methods of Aerodynamic Control.

Unit-III: Wind Power Control Strategies

Power Control Classification, Integrated Aerodynamic and Electric Control Strategies, Constant Speed Generator Fixed Pitch (CSG-FP) Configuration, Constant Speed Generator Variable Pitch (CSG-VP) Configuration, Variable Speed Generator Fixed Pitch (VSG-FP) Configuration, Variable Speed Generator Variable Pitch (VSG-VP) Configuration, Power Electronic Converters, Constant Speed and Variable Speed WPPs, Back-to-Back PEC in WPP.

Unit-IV: Classification of Wind Power Plants

Constant Speed Wind Power Plants: Type-A WPP, Type-B WPP, Equivalent Circuit of WRIG of Type-B WPP, Reactive Power in Constant Speed WPP, Features and Limitations of Constant Speed WPPs, Variable Speed Wind Power Plants, Type-C WPP, Equivalent Circuit of DFIG in Type-C WPP, Salient Features and Limitations of Type-C WPP, Type-D WPP.

Unit-V: Quality and Grid Integration Issues of Wind Power

Wind Power Impact, Local Impacts of Wind Power, Systemwide Impacts of Wind Power, Wind Power Variability, Islanding, WPP Electrical Safety and Grid, WPP Inertia, Plant Load Factor (or Capacity Factor), Capacity Credit.

WPP in the Electric Grid, Requirements of Grid Connected WPP, WPP Grid Connections, Interface Issues: Short Circuit Power Control, Reactive Power Control, Voltage Control, Operational Issues: Power System Stability, Frequency Control.

REFRENCES:

1. SirajAhmed:"Wind Energy-Theory and Practice" Second Edition, PHI Learning Pvt. Ltd, New

Delhi, 2011.
2. Garg L Johnson: "Wind Energy Systems" Prentice Hall. Inc, New Jersey,1985.
3. Mukund R Patel,"Wind and Solar Power Systems", CRC press , 2004
4. Rai G D ,"Wind Power Energy Resources", Khanna Publishers New Delhi 2002
5. Daniel Hunt V," Wind Power - A Hand Book of WECS", VAN NOSTREND CO.
6. Thomas Markvart and Luis Castaser," Practical Hand Book of Photovoltaics Elsevier Publications, UK ,2000

MTPE 031-FACTS AND CUSTOM POWER DEVICES-APPLICATION TO POWER SYSTEM

L T P 3 0 0

Course Outcomes:

CO1: Distinguish the performance of Transmission line with and without FACTS Devices.
CO2: Compare the SVC and STATCOM.
CO3: Determine the operation and control of various Static Series Compensators.
CO4: Analyze the operation and control of Unified Power Flow Controller.
CO5: Elucidate various power quality issues and how are they mitigated by various FACTS Devices.

UNIT I

Reactive power flow control in Power Systems Control of dynamic power unbalances in Power System Power flow control-Constraints of maximum transmission line loading Benefits of FACTS Transmission line compensation- Uncompensated line - Shunt compensation - Series compensation –Phase angle control.

UNIT II

Reactive power compensation – Shunt and Series compensation principles – Reactive compensation at transmission and distribution level. Static versus passive VAR compensator, Static shunt compensators: SVC and STATCOM - Operation and control of TSC, TCR and STATCOM - Compensator control Comparison between SVC and STATCOM.

UNIT III

Static series compensation: TSSC, SSSC -Static voltage and phase angle regulators – TCVR and TCPAR , Operation and Control –Applications , Static series compensation – GCSC,TSSC, TCSC , Static synchronous series compensators and their Control.

UNIT IV

SSR and its damping Unified Power Flow Controller: Circuit Arrangement, Operation and control of UPFC- Basic Principle of P and Q control, Independent real and reactive power flow control- Applications. Introduction to interline power flow controller. Modelling and analysis of FACTS Controllers – Simulation of FACTS controllers Power quality problems in distribution systems.

UNIT V

Harmonics, loads that create harmonics, modelling, harmonic propagation, series and parallel resonances, mitigationof harmonics , Passive filters, active filtering – shunt , series and hybrid and their control. Voltage swells, sags, flicker, unbalance and mitigation of these problems by power line conditioners,IEEE standards on power quality.

REFRENCES:

1. K R Padiyar, "FACTS Controllers in Power Transmission and Distribution", New AgeInternationalPublishers, 2007
2. X P Zhang, C Rehtanz, B Pal, "Flexible AC Transmission Systems- Modelling and Control", SpringerVerlag, Berlin, 2006
3. 3. N.G. Hingorani, L. Gyugyi, "Understanding FACTS: Concepts and Technology of Flexible AC Transmission Systems", IEEE Press Book, Standard Publishers and Distributors, Delhi, 2001.
4. K.S.Sureshkumar, S.Ashok ,"FACTS Controllers & Applications", E-book edition, Nalanda Digital Library, NIT Calicut,2003.
5. G T Heydt , "Power Quality", McGraw-Hill Professional, 2007
6. T J E Miller, "Static Reactive Power Compensation", John Wiley and Sons, Newyork, 1982.

MTPE 032-ELECTRIC VEHICLES

Course Outcomes:

CO1: Describe the basics of electric and hybrid electric vehicles, their architecture, technologies and fundamentals.
CO2: Analyze the use of different power electronics converters and electrical machines in hybrid electric vehicles.
CO3: Able to interpret the working of different configurations of electric vehicles and its components, hybrid vehicle configurations
CO4: Elucidate the use of different energy storage systems used for hybrid electric vehicles, their control techniques, and select appropriate energy balancing technology
CO5: Ability to understand the control and configurations of HEV charging stations.

UNIT I **L T P 3 0 0**

Review of Conventional Vehicle: Introduction to Hybrid Electric Vehicles: Types of EVs, Hybrid Electric Drive-train, Tractive effort in normal driving, Energy consumption Concept of Hybrid Electric Drive Trains.

UNIT II

Architecture of Hybrid Electric Drive Trains, Series Hybrid Electric Drive Trains, Parallel hybrid electric drive trains Electric Propulsion unit, Configuration and control of DC Motor drives, Induction Motor drives, Permanent Magnet Motor drives, switched reluctance motor.

UNIT III

Introduction to Energy Storage Requirements in Hybrid and Electric Vehicles: - Battery based energy storage and its analysis, Fuel Cell based energy storage and its analysis, Hybridization of different energy storage devices. Sizing the drive system, Design of Hybrid Electric Vehicle and Plug-in Electric Vehicle.

UNIT IV

Energy Management Strategies, Automotive networking and communication, EV and EV charging standards, V2G, G2V, V2B, V2H. Business: E-mobility business, electrification challenges, Business- E- mobility business, electrification challenges.

UNIT V

Connected Mobility and Autonomous Mobility- case study E-mobility Indian Roadmap Perspective. Policy: EVs in infrastructure system, integration of EVs in smart grid, social dimensions of EVs. Simulations andcase studies in above mentioned areas.

REFERENCES:

1. Emadi, A. (Ed.), Miller, J., Ehsani, M., "Vehicular Electric Power Systems" Boca Raton, CRC Press,2003.
2. Husain, I. "Electric and Hybrid Vehicles" Boca Raton, CRC Press, 2010.
3. Larminie, James, and John Lowry, "Electric Vehicle Technology Explained" John Wiley and Sons,2012.
4. Tariq Muneer and Irene IllescasGarcía, "The automobile, In Electric Vehicles: Prospects andChallenges", Elsevier, 2017.
5. Sheldon S. Williamson, "Energy Management Strategies for Electric and Plug-in Hybrid ElectricVehicles", Springer, 2013.

MTPE-033 POWER SYSTEM DEREGULATION

L T P 3 0 0

Course Outcomes:

CO1: Describe the need for restructuring of Power Systems, discuss different market models, different stakeholders and market power
CO2: Recognize and generalize the functioning and planning activities of ISO.
CO3: Understand transmission open access pricing issues and congestion management.
CO4: Analyze the transfer capability and estimate the transfer capability of a small power systems. (Numerical examples)
CO5: Elucidate ancillary services and understand reactive power as ancillary service and management through synchronous generator

Unit-I: Deregulation: Reconfiguring Power systems, unbundling of electric utilities, reasons for restructuring / deregulation of power industry, requirements and key issues – restructuring models – independent system operator (ISO), benefits from a competitive electricity market, after effects of deregulation, current situation in various deregulated markets in the world.

Unit-II: Fundamentals of Economics: Introduction, Consumer behavior, Supplier behavior, Market Equilibrium, Various costs of production, Perfectly competitive market, The Philosophy of Market Models: Introduction, Comparison of various market models, Market architecture.

Unit-III: Power wheeling: Transmission open access, pricing of power transactions, congestion management in deregulation, Transmission Pricing Methods and Loss Allocation Algorithms Classification of congestion management methods, calculation of ATC, nodal pricing, fundamentals of Locational Marginal Pricing (LMP), LMP formulation and implementation using DCOPF.

Unit-IV: Ancillary services: General description of some ancillary services, classification, load-generation balancing related services, voltage control and reactive power support services, black start capability services, ancillary services management in some deregulated electricity markets.

Unit-V: Reforms in Indian Power Sector: Framework of Indian Power Sector, The Availability Based Tariff (ABT) and Deviation Settlement Mechanism (DSM), Indian Electricity Act 2003, open access issues, power exchange, discussion of role of RLDC, NLDC and SLDC

REFERENCES:

1. K. Bhattacharya, MHT Bollenand J.C Doolder, "Operation of Restructured Power Systems", Kluwer Academic Publishers, USA, 2001. [Unit-I, II, III, IV&V]
2. Lei Lee Lai, "Power System restructuring and deregulation", John Wiley and Sons, UK.2001.
3. Fred I Denny and David E. Dismukes, "Power System Operations and Electricity Markets", CRCPress, LLC, 2002. [Unit-I, II, III, IV&V].

MTPE 041-SMART GRID

L T P 3 0 0

Course Outcomes:

CO-1- Analyze the information processing and communications to the power grid.
CO-2- Demonstrate and conceptualize the design of smart grid by selecting appropriate communication technologies, implementing smart meter and FACTS.
CO-3- Demonstrate the principles and requirements of the next generation future power network (or smart grid), using the latest trends in IoT for power systems.
CO-4- Start and observe advanced knowledge of electrical power system operations and control to analyze the challenges and opportunities due to increased penetration of renewable energy sources.
CO-5- Understand significance of automation and digitization in Transmission and Distribution.

UNIT I

Introduction to Smart Grid: Evolution of Electric Grid, Concept of Smart Grid, Definitions, Need ofSmart Grid, Functions of Smart Grid, Opportunities & Barriers of Smart Grid, Difference between conventional & smart grid, Concept of Resilient & Self-Healing Grid, Present development & International policies in Smart Grid. Case study of Smart Grid. CDM opportunities in Smart Grid.

UNIT II

Smart Grid Technologies: Part 1: Introduction to Smart Meters, Real Time Prizing, Smart Appliances, Automatic Meter Reading(AMR), Outage Management System(OMS), Plug in Hybrid Electric Vehicles(PHEV), Vehicle to Grid, Smart Sensors, Home & Building Automation, Phase Shifting Transformers.

UNIT III

Smart Grid Technologies: Part 2: Smart Substations, Substation Automation, Feeder Automation. Geographic Information System(GIS), Intelligent Electronic Devices(IED) & their application for monitoring & protection, Smart storage like Battery, SMES, Pumped Hydro, Compressed Air Energy Storage, Wide Area Measurement System(WAMS), Phase Measurement Unit(PMU).

UNIT IV

Micro grids and Distributed Energy Resources: Concept of micro grid, need & applications of micro grid, formation of micro grid, Issues of interconnection, protection & control of micro grid. Plastic & Organic solar cells, thin film solar cells, Variable speed wind generators, fuel cells, micro turbines, Captive power plants, Integration of renewable energy sources.

UNIT V

Power Quality Management in Smart Grid: Power Quality & EMC in Smart Grid, Power Quality issues of Grid connected Renewable Energy Sources, Power Quality Conditioners for Smart Grid, Web based Power Quality monitoring.

REFERENCES:

1. Ali Keyhani, Mohammad N. Marwali, Min Dai "Integration of Green and Renewable Energy inElectric Power Systems", Wiley
2. 2. Clark W. Gellings, "The Smart Grid: Enabling Energy Efficiency and Demand Response", CRCPress

3. JanakaEkanayake, Nick Jenkins, KithsiriLiyanage, Jianzhong Wu, Akihiko Yokoyama, "Smart Grid:Technology and Applications", Wiley
4. Jean Claude Sabonnadière, NouredineHadjsaïd, "Smart Grids", Wiley Blackwell 19
5. Peter S. Fox Penner, "Smart Power: Climate Changes, the Smart Grid, and the Future of ElectricUtilities", Island Press; 1 edition 8 Jun 2010
6. S. Chowdhury, S. P. Chowdhury, P. Crossley, "Microgrids and Active Distribution Networks."Institution of Engineering and Technology, 30 Jun 2009
7. Stuart Borlase, "Smart Grids (Power Engineering)", CRC Pre

MTPE 042-DISTRIBUTED GENERATION

L T P 3 0 0

Course Outcomes:

CO1. Describe the current scenario of Distributed Generation and the need to implement DG sources
CO2. Associate different types of micro-grids and Investigate the different types of interfaces for Grid integration of DGs.
CO3. Evaluate the various impacts of DGs on system performance and analysis of stability
CO4. Appraise and evaluate the reactive power and voltage control of DGs and various control aspects upon transmission and distribution systems.
CO5. Analyze the islanding schemes and protection related issues in micro-grids.

UNIT I

DG Definitions and Standards, DG potential

Definitions and terminologies; current status and future trends,Need of distributed generation, DG Planning

DG Technologies

DG from renewable energy sources DG from non-renewable energy sources Sitting and Sizing of DG

UNIT II

Distributed generation and Energy Storage applications

Base load; peaking; peak shaving and emergency power Need of Energy storage Batteries, Flywheels and Super capacitors

DG interconnection, Interconnection Requirements

Characteristics of DG interface: Rotating machines Characteristics of DG interface: Static power converters General protection requirement; effect of transformer connections.

UNIT III

Impact of DGs upon Technical and economical impacts Protective relaying Transient and dynamic stability of transmission line Distribution systems.

UNIT IV

Control and Reliability aspects of DGs, Voltage control techniques, Reactive power Control Power quality issues Reliability in steady state and Dynamic state.

UNIT V

Micro-Grids Concepts, types ,Sizing Modelling & Analysis Modelling & Analysis-with DG Power electronic Interfacing Transient in micro-grids Protection of Micro-grids

REFRENCES:

1. H. Lee Willis, Walter G. Scott, 'Distributed Power Generation – Planning and Evaluation', MarcelDecker Press, 2000.
2. M.GodoySimoes, Felix A.Farret, 'Renewable Energy Systems – Design and Analysis with InductionGenerators', CRC press.
3. Robert Lasseter, Paolo Piagi, 'Micro-grid: A Conceptual Solution', PESC 2004, June 2004.
4. F. Katiraei, M.R. Iravani, 'Transients of a Micro-Grid System with Multiple Distributed Energy Resources', International Conference on Power Systems Transients (IPST'05) in Montreal, Canadaon June 19-23, 2005.
5. Z. Ye, R. Walling, N. Miller, P. Du, K. Nelson, 'Facility Microgrids', General Electric GlobalResearch Center, Niskayuna, New York, Subcontract report, May 2005.

MTPE 043- MACHINE LEARNING APPLICATION TO POWER SYSTEMS

L T P 3 0 0

Course Outcomes:

CO1. Investigate the current scenario Machine learning and the comparison and relation to artificial intelligence.
CO2. Associate different types of Machine Learning algorithms and differentiate Supervised learning, unsupervised learning.
CO3. Investigate the Neural Network architectures Single layer feed forward Network and Multi layer feed forward network
CO4. Appraise and evaluate the Similarities and differences between GA and traditional methods.
CO5. Analyze the Applications of machine learning in power systems operation and control for solving problems of load forecasting.

UNIT 1
Introduction to Machine learning, History and early works, techniques, comparison and relation toartificial intelligence, optimization and statistics.

UNIT 2
Theoretical aspects of ML, different types of Machine Learning algorithms such as Linear regression, Logistic regression, K - Nearest Neighbor, Artificial Neural Networks, Random Forest, and Support Vector Machine, learning approaches: Supervised learning, unsupervised learning, semi supervised learning, reinforcement learning, self learning and association rules.

UNIT 3
Artificial Neural Network, Basic Concept, early NN Architectures, Characteristics, Neural Network architectures, Single layer feed forward Network, Multi layer feed forward network, recurrent networks,Non linear activation operators, learning methods like Back propagation, LM etc., training and testing ofANN.

UNIT 4
Genetic Algorithms: Fundamentals, History, working principal, genetic modeling, encoding, fitness function, Genetic operators: reproduction, cross over, mutation, Similarities and differences between GAand traditional methods; Unconstrained and constrained optimization using Genetic Algorithm.

UNIT 5

Applications of machine learning in power systems operation and control for solving problems of load forecasting, renewable energy forecasting, load flow studies, Economic load dispatch, Unit commitment, power plant monitoring, fault identification and security assessment etc.

REFERENCES:

1. The Elements of Statistical Learning, by Trevor Hastie, Robert Tibshirani, Jerome H. Friedman
2. Pattern Recognition and Machine Learning, by Christopher Bishop
3. NP Padhy , Artificial Intelligence and Intelligent Systems, Oxford University Press
4. Rajasekaran S. and Pai G.A.V., "Neural Networks, Fuzzy Logic and Genetic AlgorithmSynthesis and applications", PHI New Delhi.
5. Goldberg D.E. "Genetic Algorithms in Search Optimization & Machine Learning", AdditionWesley Co., New York.
6. . Kalyanmoy Deb "Optimization for Engineering Design", PHI publication.
7. Kevin Warwick, Arthur Ekwue, Raj Agrawal "Artificial intelligence techniques in power.

MTPE 051 POWER SYSTEM DYNAMICS AND CONTROL

L T P 3 0 0

Course Outcomes:

CO-1- To develop dynamic modeling of a synchronous machine.
CO-2- Demonstrate the modeling of excitation and speed governing system.
CO-3- To analyze the small signal stability without controllers.
CO-4- To analyze the small signal stability with controllers.
CO-5- Elucidate the methods to enhance the small signal stability of the power system.

UNIT – I

Introduction to Power System Stability: Power System Operation and Control, Stability Problems faced by Power Systems, Impact on Power System Operation and Control.

Analysis of Dynamical Systems :Concept of Equilibria, Small and Large Disturbance Stability, Example: Single Machine Infinite Bus System, Modal Analysis of Linear Systems, Analysis using Numerical Integration Techniques, Issues in Modelling: Slow and Fast Transients, Stiff Systems.

UNIT – II

Modeling of a Synchronous Machine Physical Characteristics, Rotor Position Dependent model. D-Q Transformation, Model with Standard Parameters, Steady State Analysis of Synchronous Machine, Short Circuit Transient Analysis of a Synchronous Machine. Synchronous Machine Connected to Infinite Bus.

UNIT – III

Modelling of Excitation and Prime Mover Systems Physical Characteristics and Models, Control system components, Excitation System Controllers, Prime Mover Control Systems., Modelling of

Transmission Lines and Loads Transmission Line Physical Characteristics, Transmission Line Modelling, Load Models - induction machine model., Other Subsystems - HVDC, protection systems.

UNIT-IV

Stability Issues in Interconnected Power Systems Single Machine Infinite Bus System,Multi-machine Systems. Stability of Relative Motion, Frequency Stability: Centre of Inertia Motion. Concept of Load Sharing: Governors. Single Machine Load Bus System: Voltage Stability. Torsional Oscillations.

UNIT – V

Power System Stability Analysis Tools Transient Stability Program, Small Signal Analysis Program, EMTP Programs, Real-Time Simulators.

Enhancing System Stability Planning Measures, Stabilizing Controllers (Power System Stabilizers)., Operational Measures- Preventive Control. Emergency Control.

REFRENCES:

1. K.R.Padiyar, Power System Dynamics, Stability & Control, 2nd Edition, B.S. Publications,Hyderabad, 2002.
2. P.Kundur, Power System Stability and Control, McGraw Hill Inc, New York, 1995.
3. P.Sauer&M.A.Pai, Power System Dynamics & Stability, Prentice Hall, 1997..

MTPE 052-MODERN POWER SYSTEM PROTECTION

L T P 3 0 0

Course Outcomes:

CO1: Describe the important operating principle, design and planning of the protective system in a power system.
CO2: Determine the traditional electro-mechanical relaying principle as well as to modern numerical relaying basics.
CO3: Describe the working principle and basic components of wide area measurement systems.
CO4: To analyze the concept of synchronizing phase measurement unit.
CO5: To analyze the concept of auto reclosing.

Unit-I: Introduction to Digital Protection Relays-1

Classification of protective relays, Electro-mechanical solid state and digital relays, adoptive relaying, tripping mechanism of relay, different relay algorithms, Comparison of digital relays with previous generation relays, Basic Components of Digital Relays with block diagram, Signal Conditioning Subsystems, Surge Protection Circuits, Anti aliasing filter.

Unit-II: Introduction to Digital Protection Relays-2

Conversion Subsystem, The Sampling Theorem, Sample and Hold Circuit, Concept of analog to digital and digital to analog conversion, Idea of sliding window concept, introduction to intelligent electronic device (IED), Fourier analysis based half cycle and full cycle algorithm

Unit-III: Measurement for Protection and Coordination of Relays in an Interconnected Power System Network

Protection of an interconnected system, Flowchart of Primary/Backup relay pairs, Flowchart of Time Multiplier Setting. Examples based on existing power system network

Protective Current & Potential Transformers: Different Types, Accuracy, burden, Polarity and Transient Response.

Unit-IV: Auto-reclosing and Synchronizing

Introduction, advantageous of auto-reclosing, classification of auto-reclosing, Sequence of events in single-shot auto-reclosing scheme, dead time, reclaim time, instantaneous trip lockout, intermediate lockout, breaker supervision function, Synchronism check, phasing voltage method, angular method, automatic synchronization.

Unit-V: Microprocessor Based Relaying

Microprocessor Based Protective Relays: (Block diagram and flowchart approach only) Over current relays ,impedance relays

Wide Area Protection and Measurement

Definition of wide-area protection, Architectures of wide-area protection, concept of synchronized sampling, wide area phasor measurement technology.

REFERENCES:

1.Badri Ram and D. N. Vishwakarma, “Power system protection and Switch gear “, TMH publication New Delhi

2. T.S. Madhav aRao , “Static relays”, TMH publication,
3. Digital Protection- Protective Relaying from Electromechanical to Microprocessor By L. P. Singh, New Age International
4. Protection and Switchgear, Bhavesh Bhalja, R. P. Maheshwari, Nilesh G. Chothani, Oxford University Press.
5. Power System Protection By Patra Basu & Choudhary, Oxford & IBH

MTPE 053-SYNCHROPHASORS APPLICATIONS TO POWER SYSTEM

L T P 3 0 0

Course Outcomes:

CO1.Describe the fundamentals and measurement principles of synchropahsors.
CO2 Analyze the Synchrophasor metrics and voltage sensitivity related to synchrophasors.
CO3.Analyze the Importance of PMU location with the help of 8 bus reference system.
CO4.To identify event type using system frequency signature and different types of grid event signatures.
CO5.To identify and analyze oscillations and monitoring the oscillations.

UNIT-I

SYNCHROPHASOR FUNDAMENTALS: Fundamentals, Definition and Description, Synchrophasor Attributes – Essential Properties, Applying Synchrophasors – Measurement Principles and Examples, Synchrophasor Technology Infrastructure – Components and Data Flow.

UNIT-II

SYNCHROPHASOR METRICS – USE IN REAL TIME OPERATIONS: Introduction to Synchrophasor Metrics, Phase Angle Differences, Voltage Sensitivity, Frequency Deviations, Oscillations.

UNIT-III

PHASE ANGLE DIFFERENCES: Definition & Importance of Phase Angle Differences for Operations, Importance of PMU Location, Use of Phase Angles in Control Rooms - Line Closing, Case Study – 8 Bus System, Use of Phase Angles in Control Rooms - Islanding & System Separation, Phasor Assisted Line Reclosing.

UNIT-IV

GRID EVENT SIGNATURES – USE IN OPERATIONS TO DETECT & DIAGNOSE GRID EVENTS : Introduction to Grid Event Signatures, Types of Grid Event Signatures - Generation Trip, Line Fault, Line Trip, Load Trip, Islanding, Oscillations, Identify Event Type Using System Frequency Signature, Event Diagnosis - Using Synchrophasor Metrics, Case Study – Generation Trip and Line Trip.

UNIT V

POWER SYSTEM OSCILLATIONS – TYPES, CAUSES, MONITORING: Introduction to Power System Oscillations, Identifying and Analysing Oscillations, Oscillation Monitoring, Oscillation Detection,

REFERENCES:

1. A. G. Phadke and J. S. Thorp, Synchronized Phasor Measurements and their Applications, Springer, 2008.
2. IEEE Transactions, Special Issue in Proceedings of the IEEE and IET proceedings.
3. IEEE standard C37-118, and other standards.

MTPE 251-POWER ELECTRONICS LAB

Course Outcomes: **L T P 0 0 2**

CO-1. Design the control circuit and the power circuit for DC-DC converters.
CO-2. Verify the compliance of spectral performance of a Three-phase voltage source Inverter.
CO-3. Critically compare various options available for the drive circuit requirements.
CO-4. Analyze and to understand the characteristics of half and fully controlled converter. Simulate Single phase and three phase converters and voltage controllers.
CO-5. Analyze and to understand the characteristics of AC voltage Controller.

Experiments and computer simulations on:

1. Experimental study for characteristics of single phase semi controlled Full Bridge converter.
2. Experimental study for single phase AC voltage controller using TRIAC.
3. Experimental study for characteristics of single phase fully controlled Full Bridge converter.
4. Experimental study for characteristics of three phase semi controlled Full Bridge converter.
5. Experimental study for characteristics of three phase fully controlled Full Bridge converter.
6. Experimental study for characteristics of DC-DC Buck converter.
7. Experimental study for characteristics of DC-DC Boost converter.
8. Experimental study for characteristics of DC-DC Buck-Boost converter.
9. Experimental study for single phase Inverter.
10. Experimental study for three phase Inverter.

In addition to the above, the Department can offer a few other experiments Minimum of 10 experiments are to beconducted.